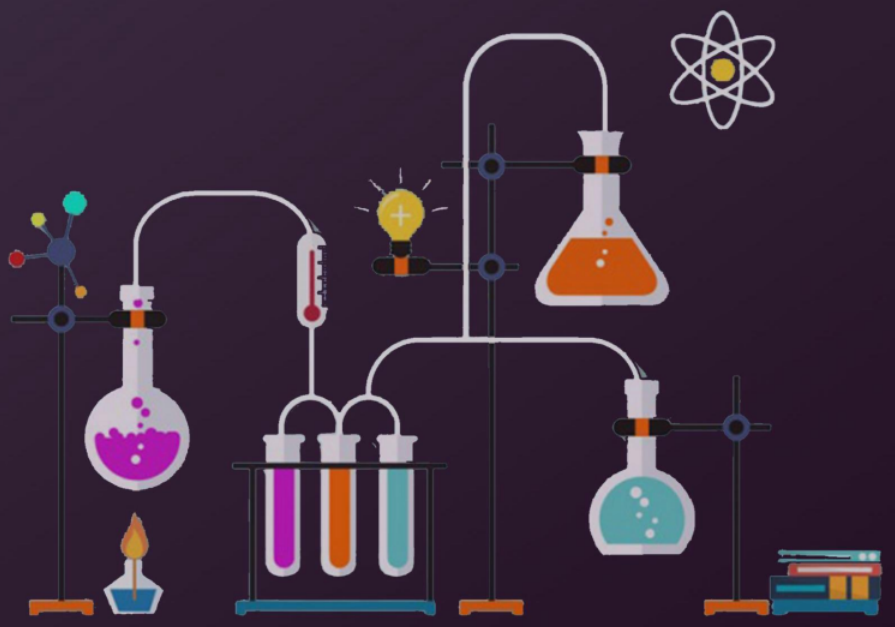
CHEMISTRY

Nuclear Chemistry



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

S.No	Topic Name	Ex 1	Ex 1A	Ex 2	Ex 2A
1	Introduction to Nuclear chemistry	1			
2	Nuclear Force Theory		1,2,8,9	1	
3	Binding Energy and Mass Defect	2-6	3,4,5,7	2,3	3,4
4	Q value of nuclear reaction	7,8		4,5	1,2,5,6
5	Which nucleus is more stable?	9	6		
6	Life lesson from einstein				
7	Einstein and his world's most famous equation				

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction to Nuclear Chemistry

Q1. Hydrogen Bomb is based on the Phenomenon of-

(A) Nuclear Explosion

(B) Chemical Reaction

(C) Nuclear Fusion

(D) Nuclear Fission

Binding Energy and Mass Defect

Q2. The value of energy for ${}_1H^2$, ${}_2He^4$, ${}_{26}Fe^{56}$, ${}_{92}U^{235}$ are 2.22, 28.3, 492 and 1786 units respectively. Then most stable nucleus is-

(A) ${}_{26}Fe^{56}$ (B) ${}_1H^2$

(C) ${}_{92}U^{235}$ (D) ${}_2He^4$

Q3. How much energy will be released when 10 kg of U^{235} is completely converts into energy :-

(A) 5×10^{27} MeV (B) 5×10^{24} MeV

(C) 9×10^{17} J (D) All of these

Q4. A g ray photon produces an electron positron pair, If the rest mass energy of electron is 0.51 Mev and the total kinetic energy of electron - positron pair is 0.78 MeV then the energy of γ - ray photon in Mev is :-

(A) 0.78 (B) 1.8

(C) 1.28 (D) 0.28

Q5. Let u be denote one atomic mass unit. One atom of an element of mass number A has mass exactly equal to Au

(A) for any value of A

(B) only for A = 1

(C) only for A = 12

(D) for any value of A provided the atom is stable

Q6. The binding energy per nucleon for C^{12} is 7.68 MeV and that for C^{13} is 7.5 MeV. The energy required to remove a neutron from C^{13} is

(A) 5.34 MeV (B) 5.5 MeV

(C) 9.5 MeV (D) 9.34 MeV

Q value of a Nuclear Reaction

Q7. Binding energy per nucleon of deutron is 1.112 MeV and binding energy per nucleon of an α-particle is 7.07 MeV, then in following process energy Q is :–

$2({}_1H^2) \rightarrow {}_2He^4 + Q$

(A) 1 MeV (B) 11.9 MeV

(C) 23.8 MeV (D) 931 MeV

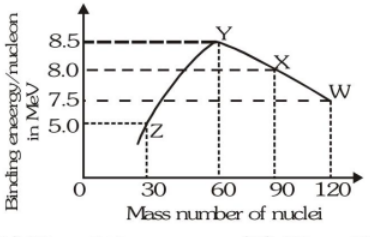
Q8. The nucleus of element X (A = 220) undergoes α-decay. If Q-value of the reaction is 5.5 MeV, then the kinetic energy of α-particle is :

(A) 5.4 MeV (B) 10.8 MeV

(C) 2.7 MeV (D) 11.2 MeV

Which Nucleus is more stable?

Q9. Binding energy per nucleon versus mass number curve for nuclei is shown in figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would release energy is :



(A) $Y \rightarrow 2Z$ (B) $W \rightarrow X + Z$

(C) $W \rightarrow 2Y$ (D) $X \rightarrow Y + Z$

Exercise - 1A
Concept Builder

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If a nucleus is emitting e^- particle, its neutron to proton ratio $\left[\frac{n}{p}\right]$ will :–

(A) increase
(B) decrease
(C) remain unchanged
(D) can't be determined

Q2. If radius of the $^{27}_{13}Al$ nucleus is estimated to be 3.6 fermi, then the radius of $^{125}_{52}Te$ nucleus be nearly-

(A) 6 fermi (B) 8 fermi
(C) 4 fermi (D) 5 fermi

Q3. The binding energies of nuclei X and Y are E_1 and E_2 respectively. Two atoms of X fuse to give one atom of Y and an energy Q is released. Then:

(A) $Q = 2E_1 - E_2$ (B) $Q = E_2 - 2E_1$
(C) $Q = 2E_1 + E_2$ (D) $Q = 2E_2 + E_1$

Q4. The binding energies of the atom of elements A & B are E_a & E_b respectively. Three atoms of the element B fuse to give one atom of element A. This fusion process is accompanied by release of energy e. Then E_a, E_b are related to each other as

(A) $E_a + e = 3E_b$
(B) $E_a = 3E_b$
(C) $E_a - e = 3E_b$
(D) $E_a + 3E_b + e = 0$

Q5. The binding energy per nucleon of deuteron (^2_1H) and helium nucleus (^4_2He) is 1.1 MeV and 7 MeV respectively. If two deuteron nuclei react to form a single helium nucleus, then the energy released is-

(A) 13.9 MeV (B) 26.9 MeV
(C) 23.6 MeV (D) 19.2 MeV

Q6. The instability of the nucleus can be due to various causes. An unstable nucleus emits radiations if possible to transform into less unstable state. Then the cause and the result can be

(A) a nucleus of excess nucleons is α active
(B) an excited nucleus of excess protons is β^- active
(C) an excited nucleus of excess protons is β^+ active
(D) an nucleus of excess neutrons is β^- active

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q7. Calculate the binding energy of $_{17}Cl^{35}$ (in MeV)if mass of $_{17}Cl^{35}$ nucleus is 34.98 amu, mass of neutron is 1.008665 amu, and mass of proton is 1.007277 amu.

Q8. Carbon has two stable isotopes. Natural carbon has 98.9% carbon–12 and 1.1% carbon–13, calculate the average atomic weight of carbon.

SUBJECTIVE TYPE

Q9. Two nuclei have their mass numbers in ratio 1 : 3. What is the ratio of nuclear densities ?

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. Let F_{pp}, F_{pn} and F_{nn} denote the nuclear force between proton–proton, proton–neutron and neutron–neutron pair respectively. When separation is 1 fm :-

(A) $F_{pp} < F_{pn} = F_{nn}$

(B) $F_{pp} > F_{pn} = F_{nn}$

(C) $F_{pp} = F_{pn} = F_{nn}$

(D) $F_{pp} < F_{pn} < F_{nn}$

Q2. If a star can convert all the He nuclei completely into oxygen nuclei. The energy released per oxygen nuclei is : [Mass of the nucleus is 4.0026 amu and mass of oxygen nucleus is 15.9994]

(A) 7.6 MeV (B) 56.12 MeV

(C) 10.24 MeV (D) 23.4 MeV

Q3. The mass of proton is 1.0073 u and that of neutron is 1.0087 u (u = atomic mass unit). The binding energy of $_2{}^4He$ is :–

[Given helium nucleus mass $\approx 4.0015u$]

(A) 0.0305 J (B) 0.0305 erg

(C) 28.41 MeV (D) 0.061 u

Q4. Per nucleon energy of $_3Li^7$ and $_2He^4$ nucleus is 5.60 MeV and 7.06 MeV, then in $_3Li^7 + {}_1P^1 \rightarrow 2{}_2He^4$ energy released is :–

(A) 29.6 MeV (B) 2.4 MeV

(C) 8.4 MeV (D) 17.28 MeV

Q5. In the options given below, let E denote the rest mass energy of a nucleus and n a neutron. The correct option is :

(A) $E(^{236}_{92}U) > E(^{137}_{53}I) + E(^{97}_{39}Y) + 2E(n)$

(B) $E(^{236}_{92}U) < E(^{137}_{53}I) + E(^{97}_{39}Y) + 2E(n)$

(C) $E(^{236}_{92}U) < E(^{140}_{56}Ba) + E(^{94}_{36}Kr) + 2E(n)$

(D) $E(^{236}_{92}U) < E(^{140}_{56}Ba) + E(^{94}_{36}Kr) + 2E(n)$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Consider the following nuclear reactions and select the correct statements from the options that follow.

Reaction I : $n \rightarrow p + e^- + \bar{v}$

Reaction II : $p \rightarrow n + e^+ + v$

(A) Free neutron is unstable, therefore reaction I is possible

(B) Free proton is stable, therefore reaction II is not possible

(C) Inside a nucleus, both decays (reaction I and II) are possible

(D) Inside a nucleus, reaction I is not possible but reaction II is possible.

Q2. A nuclear transformation is denoted by $X(n, \alpha) \rightarrow {}^{7}_{3}Li$. Which of the following is the nucleus of element X ?

(A) ${}^{12}_{6}C$ (B) ${}^{10}_{5}B$

(C) ${}^{9}_{5}B$ (D) ${}^{11}_{4}Be$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q3. In the following, column I lists some physical quantities & the column II gives approx. energy values associated with some of them. Choose the appropriate value of energy from column II for each of the physical quantities in column I and write the corresponding letter A, B, C etc. against the number (i), (ii), (iii), etc. of the physical quantity in the answer book. In your answer, the sequence of column I should be maintained.

Column I	**Column II**
(i) Energy of thermal neutrons	(A) 0.025 eV
(ii) Energy of X-rays	(B) 0.5 eV
(iii) Binding energy per nucleon	(C) 3 eV
(iv) Photoelectric threshold of metal	(D) 20 eV
	(E) 10 keV
	(F) 8 MeV

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q4. Find out the mass of Uranium required per day to generate 10 MW power from the fission of $_{92}U^{235}$.

SUBJECTIVE TYPE

Q5. Suppose that the Sun consists entirely of hydrogen atom and releases the energy by the nuclear reaction, $4\,{}_1^1H \longrightarrow {}_2^4He$ with 26 MeV of energy released. If the total output power of the Sun is assumed to remain constant at 3.9×10^{26} W, find the time it will take to burn all the hydrogen. Take the mass of the Sun as 1.7×10^{30} kg.

Q6. The Q value of a nuclear reaction

$A + b \rightarrow C + d$ is defined by

$Q = [m_A + m_b - m_C - m_d]c^2$

where the masses refer to the respective nuclei. Determine from the given data the Q-value of the following reactions and state whether the reactions are exothermic or endothermic.

(i) ${}_1^1H + {}_1^3H \rightarrow {}_1^2H + {}_1^2H$

(ii) ${}_6^{12}C + {}_6^{12}C \rightarrow {}_{10}^{20}Ne + {}_2^4He$

Atomic masses are given to be

$m({}_1^2H) = 2.014102$ u

$m({}_1^3H) = 3.016049$ u

$m({}_6^{12}C) = 12.000000$ u

$m({}_{10}^{20}Ne) = 19.992439$ u

Answer Key

Ex - 1

1	2	3	4	5	6	7	8	9
C	A	C	B	C	A	C	A	C

Ex-1A

1. B **2.** A **3.** B **4.** C
5. C **6.** A,C,D **7.** 278.36 MeV **8.** 12.011 u
9. 1:1

Ex. - 2

1	2	3	4	5
C	C	C	D	A

Ex-2A

1. A,B,C **2.** B **3.** (i) A, (ii) E, (iii) F, (iv) C **4.** 10.5 g
5. $8/3 \times 10^{18}$ sec **6.** (i) –4.054 MeV, endothermic (ii) 5.64 MeV, exothermic

-: Important Notes & Formulas :-